MAINCRAFTS TECHNOLOGY

Python Programming Internship — Task 1

Core Python Challenges | 8 Programs with Sample Notes

1. Sum of Two Numbers

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

total = a + b

print("Sum =", total)
```

Note: Example: 10 and 20 produce 30.

2. Odd or Even Checker

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

Note: The remainder operator (%) checks whether the number is divisible by 2.

3. Factorial Calculation

```
num = int(input("Enter a non-negative number: "))

if num < 0:
    print("Factorial is not defined for negative numbers")
else:
    fact = 1
    for i in range(1, num + 1):
        fact *= i

    print("Factorial =", fact)
```

Note: Example: 5! = 5 × 4 × 3 × 2 × 1 = 120. The factorial of 0 is 1.

4. Fibonacci Sequence

```
n = int(input("Enter the number of terms: "))

a = 0
b = 1

for i in range(n):
    print(a, end=" ")
    a, b = b, a + b

print()
```

Note: Example: For 7 terms, the output is 0 1 1 2 3 5 8.

5. String Reverse

```
text = input("Enter a string: ")

reverse_text = text[::-1]

print("Reversed string:", reverse_text)
```

Note: Example: HELLO becomes OLLEH.

6. Palindrome Check

```
text = input("Enter a word: ")

if text == text[::-1]:
    print("Palindrome")
else:
    print("Not a palindrome")
```

Note: Example: MADAM is a palindrome because it reads the same forward and backward.

7. Leap Year Check

```
year = int(input("Enter a year: "))

if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print("Leap year")
else:
    print("Not a leap year")
```

Note: A year is a leap year if it is divisible by 400, or divisible by 4 but not by 100.

8. Armstrong Number

```
num = int(input("Enter a non-negative integer: "))

if num < 0:
    print("Please enter a non-negative integer")
else:
    temp = num
    power = len(str(num))
    total = 0

    while temp > 0:
        digit = temp % 10
        total += digit ** power
        temp //= 10

    if total == num:
        print("Armstrong number")
    else:
        print("Not an Armstrong number")
```

Note: Example: 153 is an Armstrong number because $1^3 + 5^3 + 3^3 = 153$. This program also treats 0 as an Armstrong number.